

VASTMARKUM, Sweden

WMO: 022600

Lat: 63.1243N

Lon: 18.2518E

Elev: 143

StdP: 99.62

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.1	-25.2	0.4	-21.2	-22.2	0.5	-18.9	7.3	0.4	6.5	-0.9	0.8	310	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	24.9	18.1	23.0	16.6	21.2	15.7	19.0	23.1	17.7	21.4	16.6	19.9	2.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.7	20.6	16.3	11.8	19.2	15.2	11.0	18.1	54.4	23.0	50.5	21.4	47.1	19.8	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.7	5.0	DB	-25.6	27.1	3.2	2.3	-27.9	28.8	-29.7	30.1	-31.5	31.4	-33.8	33.0	(4)
(5)			WB	-25.3	20.3	2.6	1.5	-27.2	21.4	-28.7	22.2	-30.2	23.0	-32.0	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-7.0	-7.0	-3.2	1.8	7.2	11.4	14.7	13.1	8.7	2.9	-1.2	-4.9	(6)	
	DBStd	8.63	5.74	6.14	4.98	3.30	3.51	2.85	2.74	2.77	3.04	3.49	4.28	5.70	(7)	
	HDD10.0	2861	528	475	409	244	101	17	1	6	60	219	337	463	(8)	
	HDD18.3	5563	787	708	668	494	343	207	117	163	290	477	587	722	(9)	
	CDD10.0	344	0	0	0	0	15	60	147	102	20	0	0	0	(10)	
	CDD18.3	6	0	0	0	0	0	0	5	1	0	0	0	0	(11)	
	CDH23.3	136	0	0	0	0	3	21	91	21	0	0	0	0	(12)	
	CDH26.7	14	0	0	0	0	0	2	12	1	0	0	0	0	(13)	
(14) Wind	WSAvg	2.0	1.8	1.9	2.2	2.0	2.2	2.3	1.9	1.7	1.8	1.9	1.8	1.9	(14)	
(15) Precipitation	PrecAvg	745	62	38	42	42	37	56	90	92	71	80	76	60	(15)	
	PrecMax	1233	125	89	101	121	75	140	223	166	197	226	258	118	(16)	
	PrecMin	488	7	7	6	5	0	11	9	17	17	14	22	4	(17)	
	PrecStd	154	32	20	22	29	19	28	49	49	47	52	57	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	7.2	11.0	17.3	23.4	26.3	28.2	25.7	19.6	13.2	9.4	6.5	(19)	
		MCWB	4.0	3.7	6.1	10.2	14.4	15.5	19.1	18.5	14.8	10.5	7.8	4.7	(20)	
	2%	DB	4.1	5.4	7.9	13.7	20.2	22.9	25.8	23.4	17.5	11.5	7.1	4.5	(21)	
		MCWB	2.8	3.1	4.1	8.2	13.3	14.7	18.2	17.6	13.5	9.6	6.3	3.1	(22)	
	5%	DB	2.4	3.4	6.1	10.7	17.7	20.7	23.7	21.6	16.1	10.1	5.5	3.3	(23)	
		MCWB	1.2	1.7	2.9	6.0	11.8	13.8	17.6	16.7	12.9	8.7	4.9	2.3	(24)	
	10%	DB	0.7	1.4	4.2	8.6	15.3	18.8	21.9	19.7	14.5	8.9	4.5	2.1	(25)	
		MCWB	0.1	0.4	1.7	4.5	10.0	13.3	16.6	15.8	12.3	7.7	3.9	1.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.2	6.2	10.6	15.7	17.3	20.7	20.4	16.1	11.4	8.3	5.0	(27)	
		MCDB	5.9	6.4	10.3	16.3	21.7	22.1	24.7	23.5	18.3	12.3	9.1	6.2	(28)	
	2%	WB	2.8	3.3	4.5	8.4	14.0	16.1	19.5	18.7	14.6	10.1	6.5	3.6	(29)	
		MCDB	4.0	5.2	7.6	13.4	19.2	20.7	23.9	22.2	16.3	11.0	7.0	4.4	(30)	
	5%	WB	1.5	2.0	3.1	6.4	12.1	15.0	18.4	17.4	13.6	9.1	5.1	2.4	(31)	
		MCDB	2.3	3.3	5.7	10.1	16.9	19.4	22.6	20.2	15.2	10.0	5.4	3.1	(32)	
	10%	WB	0.3	0.6	1.8	4.9	10.6	13.7	17.2	16.3	12.7	7.8	4.1	1.5	(33)	
		MCDB	0.9	1.5	4.0	7.9	14.7	17.4	21.1	18.7	14.1	8.7	4.5	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.8	9.8	10.0	11.2	11.4	11.3	10.8	9.6	7.5	5.5	6.9	(35)	
		MCDBR	8.1	9.0	11.1	14.8	16.2	15.7	14.8	14.3	12.2	9.7	6.4	7.3	(36)	
	5% WB	MCWBR	7.0	7.4	7.7	9.2	9.5	8.6	8.3	8.8	8.5	7.7	5.7	6.9	(37)	
		MCWBR	7.5	8.1	10.7	13.9	15.0	13.0	13.1	12.1	10.2	8.8	6.1	6.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.223	0.283	0.307	0.344	0.344	0.341	0.358	0.349	0.318	0.304	0.371	0.345	(40)	
		taud	2.071	2.232	2.328	2.385	2.463	2.501	2.471	2.490	2.562	2.451	2.764	2.660	(41)	
	Ebn at Noon		415	657	797	840	873	882	853	830	794	645	390	243	(42)	
		Edh at Noon	30	60	80	95	96	94	95	85	65	49	27	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.16	0.74	2.23	3.78	5.01	5.71	5.23	3.97	2.41	0.99	0.25	0.08	(44)	
	RadStd		0.02	0.14	0.25	0.35	0.53	0.37	0.52	0.33	0.26	0.17	0.05	0.01	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-122	N/A	N/A	N/A

Nomenclature: See separate page